Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



TEAM DETAILS

Kosh (कोष)

Secure Agentic Digital Banking Harness

TEAM NAME
Team Kosh

TEAM LEADER
Laxman Awate

PROBLEM STATEMENT

Secure, Regulatory-Grade Agentic AI for Digital Banking Harness — bridging:

Mitra
Conversational AI

›

Sentinel
Deterministic fraud & policy authorization

›

Kavach
Cryptographic human step-up

›

Legacy Core Banking
RPA into existing CBS

Brief about the idea — Kosh (कोष): "Bharosa, har kadam."

THE PROBLEM

Banks want AI agents that execute real operations — payments, KYC, deposits, loans, legacy CBS onboarding — not FAQ bots. Yet LLMs are probabilistic and open to prompt injection & PII leaks.

KOSH'S ANSWER

Strictly separate Reasoning (Mitra), **Risk Policy** (Sentinel), **Human Step-Up** (Kavach) and idempotent **Execution**.

01 · REASON

Mitra (मित्र)

Multilingual AI banker · Gemini 3.8 Flash

27 banking tools + 11 guided workflows

- EN, HI, MR & Hinglish chat + voice replies in under 3 s
- Visitor mode: 8 products with 50% FOIR lending checks
- OCR auto-prefills PAN & Passport in guided flows

02 · PROTECT

Zero-PII Boundary

Zero-PII & zero-figure · llm_boundary.py

0 raw PII or ₹ figures ever reach the LLM

- PAN, Aadhaar, account & card numbers become [PAN_1] tokens
- Balances become {{amt_N}}, shown only in the browser
- Every turn routed via 23 formal YAML processes

03 · AUTHORIZE

Sentinel + Kavach

Policy engine + L3 cryptographic step-up

15+ hard rules & fraud weights per action

- ALLOW, CHALLENGE or DENY before any money moves
- High-risk or >₹10k: fresh OTP bound to that exact transaction
- Money tools auto-pause 15 min if denials spike

04 · EXECUTE

STP + Legacy CBS

Straight-through ops, RPA & governance

≤₹10L instant FD when policy checks pass

- Instant FD, RD, cheque book, SI & TDS certificates
- Maker-Checker RPA into legacy CBS, DOM-verified
- OAuth 2.1 /mcp gateway, 4-eyes review, HMAC audit

Opportunities — Differentiation, Problem Solving & USP

HOW IS KOSH DIFFERENT?

EXISTING BANK BOTS	KOSH
Static FAQ bots & prompt wrappers	**Actionable banking** 27 real operations, 11 workflows, instant STP & legacy CBS RPA
Hoping the LLM follows prompt rules	**Deterministic authorization** every tool call checked by Sentinel, not by prompts
Free-form prompts, all tools always on	**Formal 23-process ontology** Versioned & hashed; prunes tools, validates state changes
Closed, single-app assistants	**Open ecosystem (/mcp)** OAuth 2.1 + PKCE; 3rd-party agents propose, never see balances

USP OF THE PROPOSED SOLUTION

FOUR-STAGE AGENTIC GUARDRAIL PIPELINE

1 Ontology Router → 2 Harnessed Toolset → 3 Sentinel Risk Engine → 4 Kavach L3 Step-Up

0
raw PAN, Aadhaar, account no. or ₹ balances seen by Gemini

100%
pass on safety & fairness required by the 304-case release gate

Plus: sentence-by-sentence CheckedRelease on every streamed reply, and an HMAC-SHA256 hash-chained audit trail.

HOW WILL IT SOLVE THE PROBLEM?

Eliminates hallucinated execution

Mitra can only propose. Money moves only on Sentinel ALLOW or a Kavach OTP bound to payee, amount & account.

Prevents fraud & mule drain

₹25k/24h cap on new payees and after phone/email change, instant mule block, velocity circuit-breakers.

Bridges modern AI & legacy core

Playwright RPA does Maker-Checker entry into decades-old CBS without REST APIs, verified by DOM read-back.

Guaranteed BFSI care standards

Deterministic P0 scripts for cyber fraud (1930, RBI 3-day rule), distress (Tele-MANAS 14416) & Ombudsman cases.

List of Features Offered by Kosh

1. Mitra AI Banker & Streaming Voice

- **Multilingual:** EN, HI, MR & Hinglish with Devanagari brand accuracy (कोष)
- **Low-latency voice:** Cloud STT v2 (chirp_2) + streaming Gemini TTS
- **Responsible sales:** 50% FOIR checks, KFS/APR disclosures, governed offers

<3 s time to first audio

2. Sentinel Fraud & Policy Engine

- **Hard gates:** KYC status, delegation scope, bill binding, idempotency, ₹1L cap
- **Mule guards:** ₹25k/24h cap on new payees or after contact change; blocklist
- **Risk scoring:** amount, payee age, velocity & session → ALLOW | CHALLENGE | DENY

15+ hard policy gates per action

3. Kavach L3 Step-Up & Privacy

- **Out-of-band OTP:** Firebase step-up bound to a SHA-256 transaction digest
- **Brute-force lockout:** 3 wrong codes lock step-up for 30 minutes
- **DPDP Act 2023:** consent toggles, 90-day transcript purge, 4-eyes data wipe

30 min lock after 3 wrong codes

4. Onboarding, OCR, V-CIP & STP

- **Vision / DocAI OCR:** PAN, Passport & address proof straight into KYC forms
- **Video KYC (V-CIP):** signed HMAC webhook + SMS/WhatsApp outbox
- **Instant STP:** FD, RD (≤₹1L), cheque book, stop cheque, SI & TDS certificates

≤₹10L instant FD via STP

5. Legacy CBS Maker-Checker RPA

- **Playwright robot:** runs AO-001...AO-010 on the legacy CBS web portal
- **DOM read-back:** name, PAN, phone, email & product code verified before submit
- **Dual role:** keys as RPA-MAKER, authorises as RPA-CHECKER to issue CIF

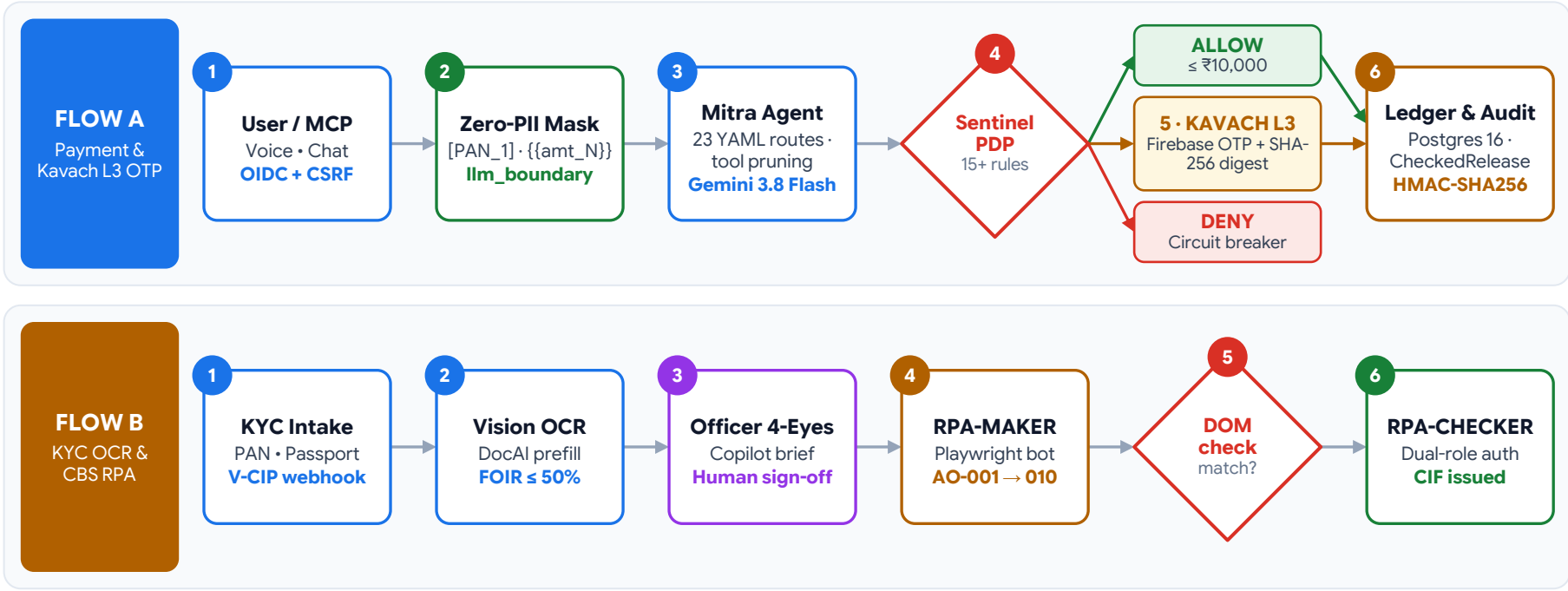
10-step CBS onboarding graph

6. Officer Copilot, MCP & Governance

- **Officer workspace:** read-only case summaries, KYC queue, AI sales-lead funnel
- **MCP gateway (/mcp):** OAuth 2.1 + PKCE + Ed25519 JWKS, propose-only
- **AI Ops:** kill switches, Ontology Copilot, Cloud DLP inspector & release gate

304-case prompt release gate

Process Flow Diagram — Agentic Payment & Legacy CBS RPA



P0 SAFETY BYPASS
care.py · 0 LLM tokens

Cyber fraud signal
→ 1930 helpline + freeze

Distress signal
→ Tele-MANAS 14416

Ombudsman grievance
→ RBI 3-day SLA flow

Wireframes & UI Surface Architecture (4 Role-Based Portals)

1 · PUBLIC PORTAL & MITRA AI DRAWER

kosh.searceinc.net · EN • HI • MR · Products • Calculators • Login

3D RuPay Hero
8 banking products

EMI / FD / RD
FOIR ≤ 50% calculator

Mitra AI Drawer
Voice STT/TTS (<3 s)
CheckedRelease

In-chat application intake
→ #/track/<ref> status tracker

2 · CUSTOMER DIGITAL BANK & KAVACH L3

kosh.searceinc.net/#/app · Session: OIDC + CSRF · Zero-PII active

14-view nav
Overview
Transfers
Bills & SI
Cards / FD
Loans / KYC

Account Ledger
{{amt_N}} UI rehydration

Kavach L3 Modal
Firebase OTP + digest

Instant STP Ops
FD / RD / card freeze

DPDP Privacy
Consent + OAuth apps

3 · BANK OFFICER WORKSPACE & COPILOT

kosh.searceinc.net/#/officer · KYC • Onboarding • High-value SRs

4-Eyes Queue
Vision OCR prefill
V-CIP webhook

Officer Copilot
Read-only brief
(no mutation)

Playwright RPA
Maker → read-back
→ Checker CIF

AI sales funnel
Offers shown → Clicked → Approved

4 · AI GOVERNANCE, SENTINEL & EVAL ADMIN

kosh.searceinc.net/#/admin · Sentinel PDP • Kill-switches • Eval gate

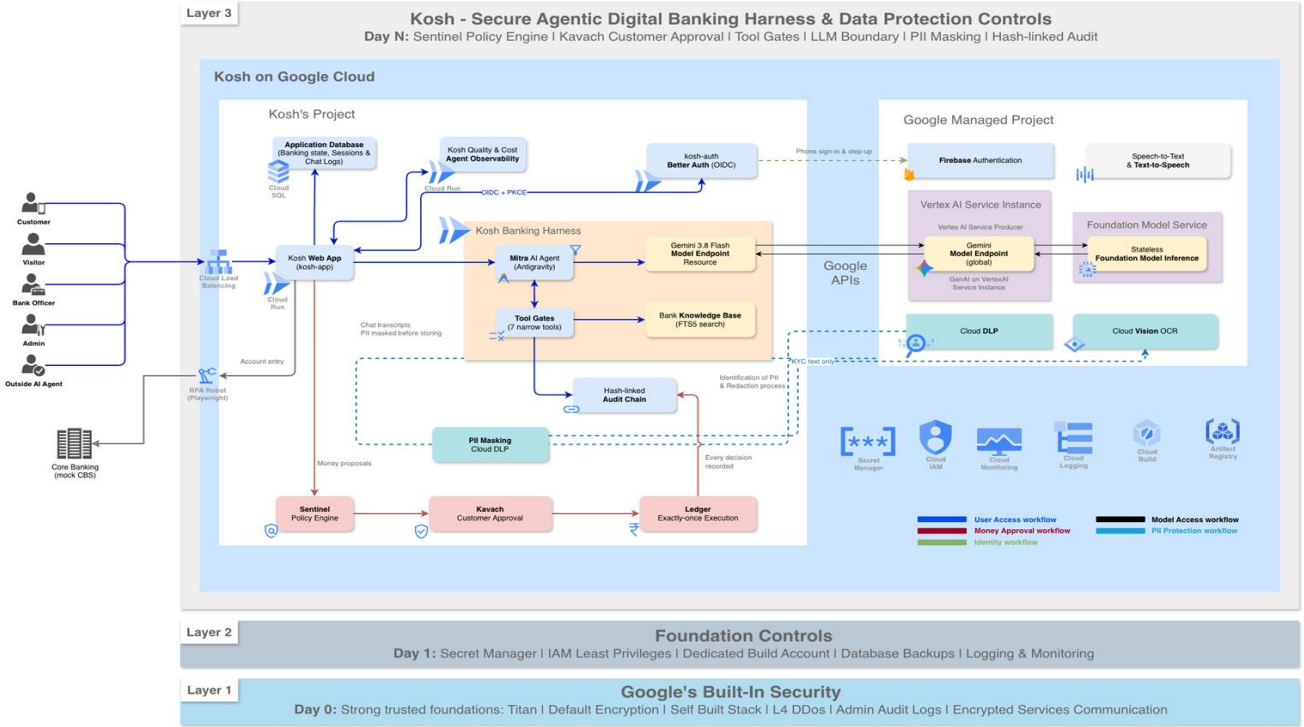
Sentinel Policy
15+ hard rules
Circuit breaker

23 YAML Ontology
Graph viewer &
SHA-256 pinning

304-Case Eval
99.7%
303/304 · 100% safety

Cloud DLP redaction tester + HMAC-SHA256 audit chain verifier

Architecture Diagram —



Click here for Zoom

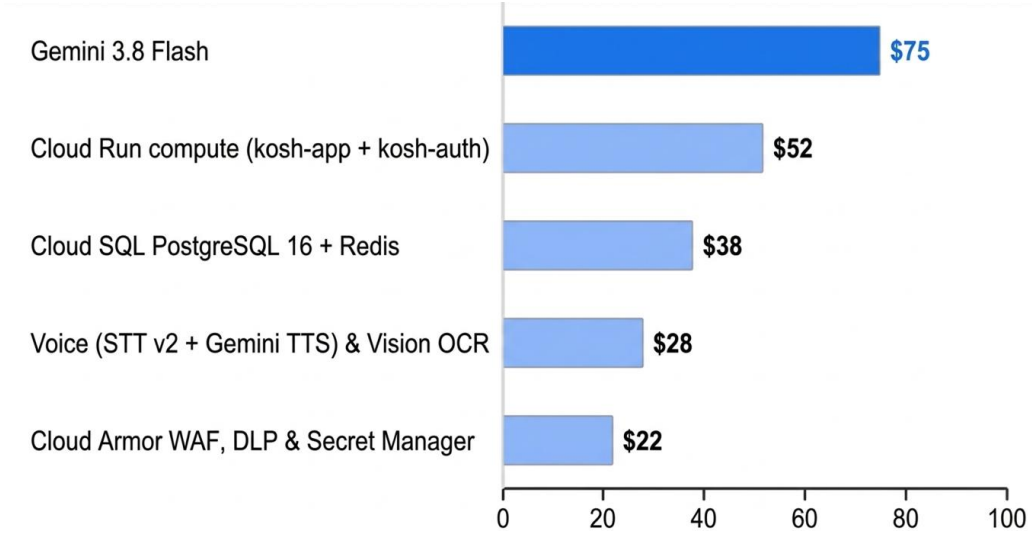
Technologies Used in Kosh

Agentic AI & Google Cloud AI	**Google Agent Development Kit** google-adk runtime, InMemoryRunner, BaseToolset & sessions	**Gemini 3.8 Flash** Primary reasoning model; auto-failover to Gemini 3.7 Flash	**Speech & Voice AI** Cloud STT v2 (chirp_2) + Gemini 3.8 Flash TTS (Kore)	**Document AI & DLP** Vision / DocAI OCR for KYC prefill + Cloud DLP
Backend, Risk Engine & Protocols	**Python 3.12 & FastAPI** Async REST + WebSocket streaming, Pydantic v2	**Sentinel & LLM Boundary** Deterministic PDP (policy.json) + Zero-PII tokenizer	**Model Context Protocol** JSON-RPC 2.0 /mcp gateway, OAuth 2.1, propose-only	**Cryptographic Audit** HMAC-SHA256 hash-chained trail across all mutations
Identity, Frontend & RPA Automation	**Better Auth OIDC** Node.js + TS OAuth 2.1 / OIDC, Ed25519 JWKS & PKCE	**Firebase Authentication** Phone OTP for sign-in (L1) & Kavach step-up (L3)	**React 19 + TypeScript + Vite** Tailwind, Framer Motion, Three.js · WCAG 2.2 AA · EN/HI/MR	**Playwright Chromium RPA** Headless worker for Maker-Checker CBS onboarding
Data Layer, DevOps & Testing	**Cloud SQL PostgreSQL 16** SQLAlchemy 2.0 + Alembic, 17 typed banking tables	**SQLite FTS5 + Redis** BM25 policy/FAQ search + rate-limiting & cache	**Cloud Run & Cloud Armor** Docker + Cloud Build microservices behind WAF	**Verification Suite** 304 LLM evals · 93 Playwright E2E · 198 pytest

Estimated Implementation & Cloud Run Cost (10,000 Active Customers)

MONTHLY COST BY COMPONENT

10,000 active customers · ~50,000 agentic turns per month



TOTAL ESTIMATED COST

$215 / mo

≈ ₹18,000 / month · production cloud-native stack

COST PER AGENTIC TURN

₹0.36 ≈ $0.0043

Across ~50,000 agentic turns per month

KEY WORKLOAD ASSUMPTIONS

- Gemini: ~60M input / 8M output tokens at $0.75 / $3.75 per 1M
- ~8,000 voice turns + ~1,500 KYC scans (PAN / Passport)
- 2 Cloud Run services (min-instances = 1) · db-f1-micro HA + 50 GB SSD

COST OPTIMIZATION BUILT INTO KOSH

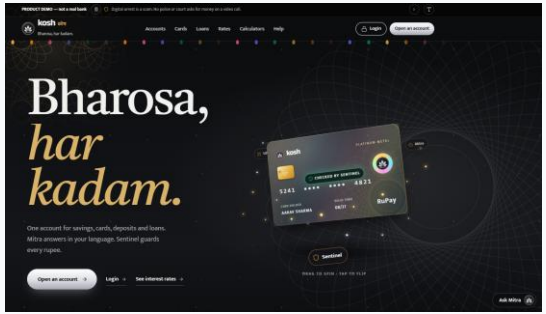
~55%
fewer prompt tokens per turn via ontology-driven tool pruning

0 extra
tool round-trips: catalogue facts embedded in visitor instructions

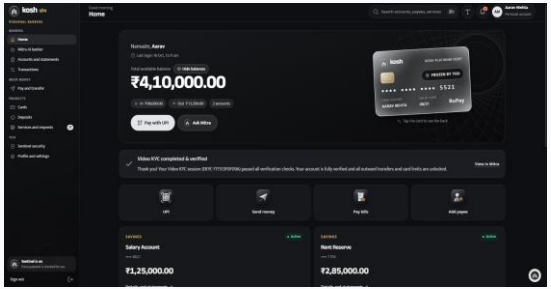
0 tokens
for deterministic P0 care / fraud scripts & STP calculators

Per-turn $
tracked live in the Admin Console via config/pricing.json

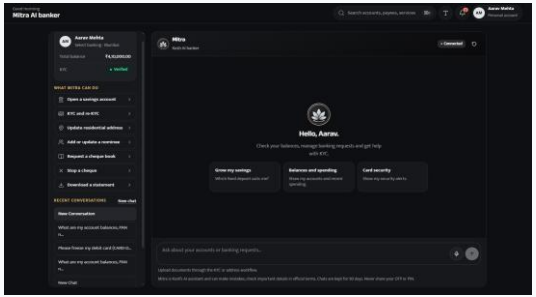
Snapshots of the Working Kosh Prototype



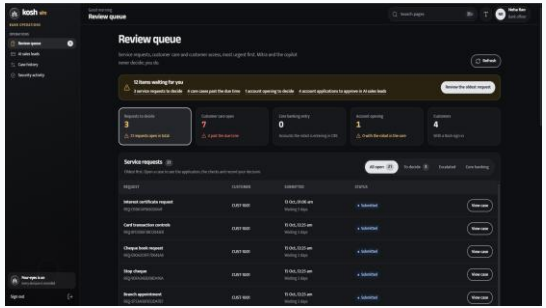
1. Public Site & Mitra AI Banker



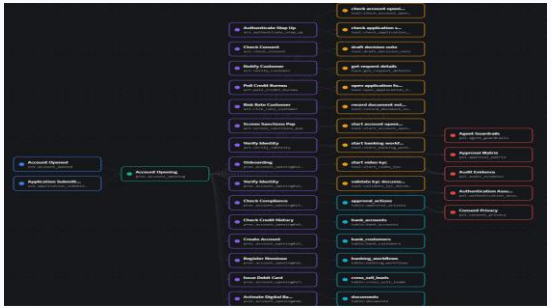
2. Customer Dashboard



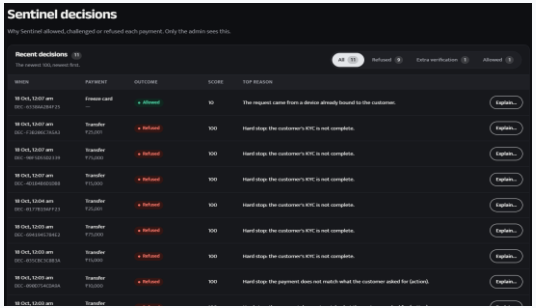
3. Mitra AI Assistant



4. Officer Dashboard View



5. Bank Ontology View



6. Sentinel Decision Dashboard

Prototype Performance Report & Benchmarking

99.7%
Live LLM evaluation pass rate
303 / 304 cases · 22 BFSI categories · Gemini 3.8 Flash

100%
Safety & fairness gate
142 / 142 · 0 jailbreaks, 0 prompt or PII leaks · 6/6 cohorts

93 / 93
Real-browser E2E suite
Full Playwright + API harness across 11 banking stages

1.2–2.9 s
Streaming voice & chat
Time to first audio · 2.3–5.0 s to first checked sentence

LLM EVALUATION SUITE — 304 CASES

tests/eval/run_eval.py · each group below passes 100%

Adversarial & jailbreak defense 27 jailbreak · 14 prompt probe · 8 customer security	49 / 49
Zero-PII & data minimisation 10 data minimisation · 25 grounding · 16 competitor guard	51 / 51
Ontology & Officer Copilot 10 process routing · 24 officer copilot · 20 signed-in	54 / 54
Vulnerable customer care (P0) 14 fraud · 9 complaint · 8 distress · 8 human · 10 lending	49 / 49
Demographic fairness cohorts Identical product picks across gender, religion & region	24 / 24

END-TO-END & ENGINEERING VERIFICATION

KOSH_TEST_REPORT.md

LATENCY GAINS before → after

First text CheckedRelease	~~8–20 s~~	2.3–5.0 s
First audio streaming TTS	~~~70 s~~	1.2–2.9 s

198 / 198
Python pytest
Unit & integration on SQLite + PostgreSQL 16

12 / 12
Frontend Vitest
Plus auth-server check

26
Defects closed
Incl. staff username takeover; service-token auth on identity endpoints

Additional Details (Regulatory Compliance) & Future Roadmap

BUILT-IN RBI & DPDP ACT 2023 COMPLIANCE

RBI digital payment security
Out-of-band L3 auth (Kavach), ₹25k/24h cooling-off for new beneficiaries & contact changes, velocity circuit-breakers

DPDP Act 2023 data governance
Banking & marketing consent toggles, 90-day transcript purge, 24h visitor memory expiry, 4-eyes data wipe

Deterministic customer protection (P0)
Hardcoded flows for cyber fraud (1930, cybercrime.gov.in, card freeze, RBI 3-day rule) & distress (Tele-MANAS 14416)

Cryptographic non-repudiation
Every tool call, Sentinel decision, officer approval & admin change in an HMAC-SHA256 hash-chained ledger

Model risk management
SHA-256 prompt/ontology pinning; release is blocked unless the running version passes the 304-case gate

FUTURE DEVELOPMENT ROADMAP

1. **Live NPCI UPI & Account Aggregator**
   Sentinel & Kavach for UPI Autopay & RuPay Credit-on-UPI; AA cash-flow underwriting for instant MSME loans
2. **Pan-India Bhashini regional voice**
   Low-latency voice banking in Tamil, Telugu, Kannada, Bengali, Gujarati & Malayalam
3. **GNN mule-ring detection**
   Real-time payment-graph scoring to catch multi-hop mule accounts across banks
4. **Pre-built enterprise CBS adapters**
   Native ISO 20022, Finacle, TCS BaNCS & Temenos connectors alongside the RPA worker
5. **On-device passkey / WebAuthn L3**
   FIDO2 biometric passkeys next to Firebase OTP for phishing-resistant Kavach signing

Project Submission Links — Kosh (कोष)

GitHub Public Repository

github.com/laxmanawate18/kosh-banking-harness

Full FastAPI backend, React 19 frontend, Better Auth OIDC server, 23-process ontology, Playwright CBS RPA worker & 304-case evaluation suite.

Open repository →

4 MIN

Demo Video Walkthrough

https://www.youtube.com/watch?v=ZfIxaeA_cuU

Visitor Mitra (EN/HI/MR + voice), KYC Vision OCR, Sentinel CHALLENGE → Kavach L3 OTP payment, Officer 4-eyes + CBS RPA, and the AI Governance Console.

Watch demo →

LIVE

Final Product (Live)

kosh.searceinc.net
auth.kosh.searceinc.net

Live production deployment on Google Cloud Run. The second link is the Identity & OAuth 2.1 server.

Open live app →

LIVE STACK
running in production

- Google Cloud Run
- Cloud SQL PostgreSQL 16
- Gemini 3.8 Flash
- External MCP Gateway (/mcp)

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

